SPRINT 2 PLANNING MEETING

Notify Engine | Algo-Rhythms | CS 691 Capstone | Pace University

Team / Application	Algo-Rhythms	**Sprint Duration**	Mar 26 – Apr 16, 2026 (3 weeks)
Application	Notify Engine	**Sprint Goal**	ML routing that outperforms static baseline

1. Sprint 2 Objective & Meeting Agenda

Goal: Integrate XGBoost-based ML routing into the notification pipeline to personalize channels by user engagement and demonstrate measurable improvement over the Sprint 1 static baseline.

- Recap Sprint 1 deliverables and confirm carry-over items
- Review backlog and select Sprint 2 stories with story points and acceptance criteria
- Assign tasks through team self-selection based on interest and availability
- Identify inter-story dependencies before starting work
- Confirm definition of done and sprint close date

2. Attendees

Name	Role	Present	Absent
Mohammed N. Yusif	Security Engineer / Developer	✓	
Krish Dhorajiya	AI Engineer / Tester	✓	
Rhythm Patel	AI Engineer / Developer	✓	
Salvatore Ardisi	Developer	✓	
Dhawalshree Mengane	Data Analyst / Developer	✓	
Zeeya Ramani	Data Analyst / Developer	✓	

3. Sprint 1 Recap

- Shipped: 10 stories, 36 story points, 0 carried over — 100% completion
- 7 working API endpoints: tenant register, send notification, status, list, summary, engagement tracking, tracking pixel
- React dashboard live with real API data — stats cards + notifications table with auto-refresh
- Full Docker Compose environment — one command starts all 6 services
- 14/14 acceptance tests passing — all criteria verified, not assumed
- Carry-over into Sprint 2: None — entering with clean slate

4. Sprint 2 Committed Backlog

ID	User Story	Acceptance Criteria	Points	Owner
PB-13	ML-powered routing — XGBoost prediction + worker integration	- POST /predict: XGBoost returns probability scores per channel - POST /train: retrain on delivery outcomes, evaluate AUC-ROC - GET /model/info: version, metrics, feature importance - 15-feature vector - Different users get different routing	**8 SP**	Salvatore
PB-14	Real-time dashboard via Socket.IO — live event feed	-routing_mode 'adaptive' triggers POST /predict - ML failure gracefully degrades to static routing - routing_decision populated on notifications table - Feature vector recorded with each attempt	**5 SP**	Zeeya, Dhawalshree
PB-15	Engagement tracking feedback loop — pixel data feeds ML training	Worker publishes to Redis pub/sub on delivery events - API subscribes and broadcasts via Socket.IO - Dashboard shows live event feed - Stats cards update in real-time - Auto-reconnect on disconnect	**3 SP**	Rhythm, Krish
PB-16	Notification fatigue detection — per-user frequency tracking	-recipient_channel_stats updated after delivery and engagement - Rolling 30d stats: attempts, successes, engagements, avg latency - ML training uses engaged flag as label - Predictions change as data accumulates	**3 SP**	Krish, Rhythm
PB-11	Rate limiting — Redis token bucket, 429 responses	-Redis token bucket per tenant_id - Default: 1000 req/min - 429 Too Many Requests on exceed - X-RateLimit-Limit, Remaining, Reset headers	**3 SP**	Yusif
PB-12	CI/CD pipeline — GitHub Actions, Jest + pytest, TruffleHog	-GitHub Actions: Jest + pytest + TruffleHog - Failing tests block merge - E2E on develop/main merges only	**2 SP**	Yusif
PB-20 (SCRUM-189)	Retro action: PR self-review checklist task	PULL_REQUEST_TEMPLATE.md in repo - Checklist: no dotenv, no any, no hardcoded creds, field names match contract"	**1 SP**	Team
PB-21 (SCRUM-191)	Retro action: Sprint 2 kickoff code walkthrough task	- 30-min session in week 1 - Cover: env loading, worker flow, dashboard API, shared types - All members attend	**1 SP**	Team
PB-19	Worker integration — adaptive routing, feature vectors, routing_decision	- routing_mode 'adaptive' triggers POST /predict - ML failure gracefully degrades to static routing - routing_decision populated on notifications table - Feature vector recorded with each attempt	**3 SP**	Salvatore
TOTAL	9 stories committed		**29 SP**	

5. Story Dependencies & Task Assignment

- PB-13 (XGBoost) depends on synthetic training data generated in Sprint 1 via PR #23 — data is ready
- Worker adaptive routing (PB-19) cannot start until ML service prediction endpoint (PB-13) is live — team to prioritize ML service integration first
- PB-14 (Socket.IO dashboard) depends on the Socket.IO server (SCRUM-182) being live — dashboard streams all delivery events, not just routing decisions
- PB-15 (engagement tracking) feeds back into PB-13 training loop- both teams must coordinate the data schema
- PB-11 (rate limiting) and PB-12 (CI/CD) are independent — can be picked up in parallel from day one

Focus Area	Tasks	Owners
ML Service (Python/Fast API)	POST /predict, POST /train, XGBoost model, epsilon-greedy routing	Salvatore
Dashboard (React)	Socket.IO live feed, routing intelligence page, fatigue indicators	Zeeya, Dhawalshree
API + Infrastructure	Rate limiting middleware, CI/CD pipeline, PR checklist template	Yusif, Krish, Rhythm

6. Definition of Done — Sprint 2

- XGBoost model trained in synthetic data and returning channel probability scores via POST /predict
- Adaptive routing live: 90% exploit highest-probability channel, 10% explore random channel
- Different users demonstrably receive different channels based on their individual engagement history
- Dashboard routing intelligence page shows channel selected, model probability, and reason
- Rate limiting enforced — requests beyond threshold return 429 with retry-after headers
- CI/CD pipeline green on every PR — Jest, pytest, and TruffleHog all passing
- PR self-review checklist template committed to repo and used on first PR

7. Risks & Mitigations

Risk	Severity	Mitigation
ML service Python container — new integration, untested in stack	**High**	Integration test against API in week 1 before other stories depend on it
Model quality depends on training data volume	**Medium**	Synthetic data already generated (PR #23) — validate size and diversity before training
Socket.IO adds real-time complexity to dashboard	**Medium**	Implement Socket.IO in isolation first, integrate with routing data after stable
Duplicate work from not reading existing codebase (retro lesson)	**Low**	30-min code walkthrough session before any Sprint 2 task is picked up

8. Meeting Close & Next Steps

- Sprint 2 goal confirmed: ML routing measurably outperforms static baseline
- Total committed: 29 story points across 9 stories
- Code walkthrough session: scheduled before first task pickup
- First standup: day after planning meeting — all blockers surfaced immediately
- Sprint review date: April 16, 2026
- All stories added to Jira Sprint 2 board with owners assigned